User Guide

What the tool shows, how to use it, and how to read the maps.

The Heat Risk Explorer is an interactive map of heat risk across San Diego County, broken down by neighborhood and by what is driving the risk in each one.

Choose which factors to include (temperature, AC access, tree cover, age, chronic illness, income, and others) and the tool builds a neighborhood-level risk map.

The tool runs entirely in the browser. Nothing is installed; nothing is sent to a server.

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Try it in 60 seconds

1. **Open the map** at connorjmack.github.io/heat-risk-viewer.
2. Leave the default variable boxes checked.
3. Click **Build Composite Risk Map**.
4. Dark red neighborhoods have the highest overall risk.
5. Click any neighborhood to see its scores and dominant driver.

The rest of the guide covers what is on the map, how the score is built, and how to use the time-period and variable controls.

Why heat hits some neighborhoods harder

A 100° day plays out differently across the county. A shaded apartment in Carlsbad with working AC and a healthy adult is one situation. A second-floor unit in El Cajon with no AC, a 90-year-old resident, and a hot roof above is another.

The tool breaks heat risk into three pieces:

- **How hot it gets:** the physical heat the neighborhood receives.
- **Who lives there:** how many residents are likely to be harmed when temperatures spike.
- **What helps people cope:** resources in the neighborhood that protect people, including AC, tree shade, and nearby healthcare.

Every neighborhood gets a score on each piece. The tool combines them into one overall heat risk score from 0 (lowest) to 100 (highest).

What goes into each piece

> ### How hot it gets shown in red
>
> The physical heat a neighborhood receives, measured at weather stations across the county:
>
> - **Daytime highs:** average summer highs, the hottest days, and the single hottest day of the year. *Best for peak heat events and short-term emergency planning, including cooling-center activation.*
> - **Nighttime lows:** how cool nights get. Bodies need cool nights to recover; when nights stay warm, heat illness rises, especially in older adults. *Best for multi-day heat waves and older-adult health, where the lack of overnight relief is what makes people sick.*
> - **Humidity:** how heavy the air is. High humidity makes the same temperature feel hotter. *Best for comparing inland and coastal neighborhoods, where the same air temperature can feel very different.*

Who lives there shown in orange

Some residents are more likely to be harmed by heat than others. The tool considers:

- **Age:** children under 18 and adults over 65. *Best for locating where heat illness will land first: senior housing, family neighborhoods, areas near schools.*
- **Health conditions:** heart disease, asthma, COPD, diabetes, high blood pressure, and other conditions that worsen in heat. *Best for healthcare partnerships, medication-safety messaging, and hospital-readiness planning.*
- **Daily-life challenges:** language barriers, disability, poverty, lack of a car, and housing-cost burden. *Best for heat communications, transit to cooling centers, and community-navigator programs.*
- **Air quality and life expectancy:** neighborhoods already living with worse air or shorter life expectancy tend to be hit harder. *Best for aligning heat work with environmental-justice or health-equity goals.*

What helps people cope shown in teal

Resources in the neighborhood that reduce harm during a heat event:

- **AC at home:** the share of homes with any AC, with central AC, or with none. *Best for cooling-center siting, utility-bill assistance, and home-AC subsidy programs.*
- **The neighborhood itself:** tree cover, parks, green space, and housing age (older homes are usually less insulated). *Best for tree-planting, park investment, cool-roof programs, and weatherization.*
- **Money and services:** income, education, health insurance, healthcare access, food access, and community connectedness. *Best for combining heat work with other safety-net programs, or for finding neighborhoods where lack of resources amplifies a heat wave's impact.*

More of these means lower heat risk. The tool accounts for that when computing the score.

How the tool puts it all together

Once you have selected your factors, the tool combines them into one score per neighborhood, scaled from 0 (lowest risk) to 100 (highest).

Two things separate this from a straight average:

Overlap is handled. Many of these factors travel together. Lower-income neighborhoods often also have older housing, less tree cover, and higher rates of chronic illness. Adding the variables directly would count the same underlying pattern several times. The tool identifies overlapping patterns and counts them once.

Protective factors reduce the score. AC, tree cover, and nearby healthcare reduce harm during a heat event. The tool subtracts those from the risk total, so a neighborhood with strong protections receives a lower score even when it gets hot.

> **Method:** the combination is done with Principal Component Analysis (PCA), a standard technique for handling overlapping variables. For a visual walkthrough of how PCA works, see this Setosa interactive.

Reading the map

The colors

Darker always means higher. On the composite map, dark red means a neighborhood is among the highest risk in the county. On the individual pillar maps, the color follows the pillar: red for heat, orange for population vulnerability, teal for protective resources.

What a score means

Scores are *relative*, not absolute. A score of 75 means the neighborhood is at higher risk than 75% of other neighborhoods in San Diego County. It does **not** mean 75% of residents will be harmed.

Clicking a neighborhood

Click any neighborhood and a panel appears with:

- Its score on each of the three pieces.
- Its overall composite score.
- Its **dominant driver:** the piece pushing its risk the highest.

The map layers

- **Composite Risk:** the overall 0–100 score combining all three pieces. The default view.
- **Dominant Driver:** each neighborhood colored by which of the three pieces matters most there. Useful for distinguishing hotspots driven by heat from those driven by population vulnerability or resource scarcity.

- **Individual pillar views:** one piece (heat, who lives there, or what helps them cope) shown on its own.

Looking into the future

Four buttons at the top of the left panel switch between time periods:

- **Present Day:** what we measure today from weather stations.
- **Moderate Emissions:** a future where emissions are cut, but not aggressively (SSP2-4.5).
- **High Emissions:** a future where emissions stay high (SSP3-7.0).
- **Very High Emissions:** a worst-case path of unchecked fossil-fuel use (SSP5-8.5).

Selecting a future scenario shifts the temperature variables using climate-model projections for 2050 or 2080. Demographics and resources stay at today's values; we do not project who will live in each neighborhood thirty years from now.

Example workflows

Find a good spot for the next cooling center

1. Start with the default selections.
2. Under **What helps people cope**, uncheck the AC variables, so the score does not credit neighborhoods that already have AC.
3. Click **Build Composite Risk Map**.
4. Look at the darkest red neighborhoods.
5. Click a few. If their dominant driver is “what helps people cope”, they are strong cooling-center candidates. If not, a cooling center alone will not address the main driver.

See which neighborhoods will get hottest by mid-century

1. Click **High Emissions** at the top of the left panel.
2. Uncheck everything except the temperature variables.
3. Click **Build Composite Risk Map**.
4. Open the tool in a second browser tab on **Present Day** to compare side-by-side. The difference shows you where warming is concentrated.

Make the case for tree planting in one neighborhood

1. Use the default selections and click **Build Composite Risk Map**.
2. Click the neighborhood of interest.
3. Note its overall score, its dominant driver, and its tree cover relative to nearby neighborhoods.
4. Screenshot the result for documentation.

Limits

The tool is not:

- A real-time heat warning. For active heat events, follow the National Weather Service.
- A statement about any individual's risk. A neighborhood score can hide the single block or household most exposed.
- A substitute for community input. The neighborhoods at highest risk are the ones whose residents should shape the response.

Common questions

Why are some neighborhoods blank on the map?

A neighborhood shows blank when one of the variables you have selected is missing for it. Unchecking the variable with the spottiest coverage will usually fill in the gaps.

Can I save my map?

Not currently; take a screenshot. The tool runs entirely in the browser, with no account and no server-side state.

Can I use this for a different county?

Not as-is. Every variable comes from a national dataset, but the data has to be assembled for each county. The method is portable; the prepared data is not.

What is the difference between “Composite Risk” and “Dominant Driver”?

Composite Risk is the overall score combining all three pieces. Dominant Driver is which of the three pieces contributes the most in a given neighborhood. Composite tells you *where* to focus; Dominant Driver tells you *what kind of solution* might help.

Who do I contact about this tool?

Open an issue at github.com/connorjmack/heat-risk-viewer/issues.

Where the data comes from

Every variable in the tool comes from a public source:

WHAT IT MEASURES	SOURCE
Weather observations	MesoWest / Synoptic Data weather station network
Future temperature projections	LOCA2 downscaled climate model output (CMIP6)
Land surface temperature	USGS Landsat satellite imagery
Chronic disease rates	CDC PLACES
Social vulnerability	CDC Social Vulnerability Index (SVI) 2022
Income, education, healthcare, housing	Healthy Places Index (HPI)
Demographics and housing	American Community Survey (5-year estimates)
Air conditioning prevalence	Modeled estimates from Romitti et al. and Ahn et al.

A small `src` link next to each variable name in the tool opens the original source in a new tab.

Glossary

Census tract

A small Census Bureau geography, usually around 4,000 people. San Diego County has roughly 630 of them.

Composite risk score

The tool's final 0–100 score combining all three pieces.

Dominant driver

The piece (heat, who lives there, or what helps them cope) that contributes the most to a neighborhood's score.

Percentile

A score from 0 to 100 indicating where a neighborhood sits relative to all others in the county. A score of 80 means higher than 80% of the others.

Climate scenario / SSP

A "what if" pathway for future emissions used in climate modeling. Higher SSP numbers mean more warming.

Principal Component Analysis (PCA)

The statistical method the tool uses to combine overlapping variables into a single score. (visual explainer)